

Calibration of the Heston Stochastic Volatility Model Using Monte Carlo Simulation and Levenberg–Marquardt Optimization

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1 Introduction

The Black–Scholes model assumes that volatility is constant. However, real option markets exhibit volatility smiles and skews, which indicates that volatility is not constant and depends on strike and maturity. This motivates stochastic volatility models, where volatility is itself random and evolves over time.

One of the most widely used stochastic volatility models is the Heston model. The goal of this paper is to calibrate the Heston model to observed market option prices. Calibration is an example of an **inverse problem**, where we attempt to recover model parameters from observed data.

2 The Heston Model

The Heston model assumes the stock price and its variance follow

$$dS_t = rS_t dt + \sqrt{v_t} S_t dW_t^{(1)}, \quad (1)$$

$$dv_t = \kappa(\theta - v_t)dt + \xi \sqrt{v_t} dW_t^{(2)}, \quad (2)$$

with correlation

$$\mathbb{E}[dW_t^{(1)} dW_t^{(2)}] = \rho dt. \quad (3)$$

2.1 Interpretation of Parameters

- v_0 : Initial variance
- θ : Long-run variance
- κ : Mean reversion speed
- ξ : Volatility of volatility
- ρ : Correlation between price and variance

3 Risk-Neutral Pricing

Under the risk-neutral measure,

$$C = e^{-rT} \mathbb{E}[(S_T - K)^+]. \quad (4)$$

4 Monte Carlo Simulation

We discretize the system as

$$v_{t+\Delta t} = v_t + \kappa(\theta - v_t)\Delta t + \xi\sqrt{v_t}\sqrt{\Delta t}Z_2, \quad (5)$$

$$S_{t+\Delta t} = S_t \exp\left((r - \tfrac{1}{2}v_t)\Delta t + \sqrt{v_t}\sqrt{\Delta t}Z_1\right). \quad (6)$$

The Monte Carlo estimator is

$$C \approx e^{-rT} \frac{1}{N} \sum_{i=1}^N (S_T^{(i)} - K)^+. \quad (7)$$

5 Calibration Problem

We minimize

$$\min \sum_i \left(\frac{C_i^{\text{mkt}} - C_i^{\text{model}}}{C_i^{\text{mkt}}} \right)^2. \quad (8)$$

6 Levenberg–Marquardt Method

We solve

$$(J^T J + \lambda I) \Delta = J^T r. \quad (9)$$

7 Calibration Results

The initial implied volatility used for initialization was

$$\sigma_{\text{ATM}} = 0.2369, \quad (10)$$

which gave

$$v_0^{\text{init}} = \theta^{\text{init}} = 0.0561. \quad (11)$$

Initial error:

$$\text{RMSE}_{\text{initial}} = 0.0762. \quad (12)$$

Final error:

$$\text{RMSE}_{\text{final}} = 0.0288. \quad (13)$$

Parameter	Value
κ	20.0000
θ	0.0550
v_0	0.0709
ξ	0.7799
ρ	-0.7061

Table 1: Calibrated Heston parameters

8 Interpretation

The parameter κ reached its upper bound, suggesting very fast mean reversion and potential identifiability issues.

The initial variance exceeds long-run variance:

$$v_0 > \theta, \tag{14}$$

indicating elevated short-term volatility.

The volatility of volatility is high:

$$\xi = 0.7799, \tag{15}$$

which generates curvature in option prices.

The correlation is strongly negative:

$$\rho = -0.7061, \tag{16}$$

capturing the leverage effect and producing skew.

9 Inverse Problem

Calibration is an inverse problem where multiple parameter sets can produce similar prices. The fact that κ reaches a boundary suggests the problem is ill-conditioned.

10 Conclusion

The calibration significantly improved fit, reducing RMSE from 0.0762 to 0.0288. While the model captures key market features such as skew and curvature, parameter identifiability remains a challenge, highlighting the importance of viewing calibration as both a numerical and statistical problem.